

Simulation study: mixture of experts model to check identifiability

The Tien Mai & Zhi Zhao

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Data simulation:

- Sample size $n = 200$
- Dimension of the Wishart distribution $p = 2$
- Number of latent components $K = 2$
- Probabilities of subpopulations $\boldsymbol{\pi} = (0.8, 0.2)$
- Degrees of freedom $\boldsymbol{\nu} = (3, 12)$
- Scale matrices of the Wishart distribution Σ_1, Σ_2
- Data $S_i \sim \pi_1 \text{Wishart}(\nu_1, \Sigma_1) + \pi_2 \text{Wishart}(\nu_2, \Sigma_2)$

Modeling:

- Bayesian mixture of experts of two Wishart distributions

```
rm(list = ls())

library(moewishart)

set.seed(123)
n <- 200 # number of subjects
p <- 2

dat <- simData(n, p,
  pis = c(0.8, 0.2),
  nus = c(3, 12),
  Sigma = NULL # default Sigmas pre-defined in function 'simData()'
)
S_list <- dat$S
Sigma_list <- dat$Sigma_list

my_K_intial <- 2

# run a moewishart model
set.seed(123)
fit <- moewishart(S_list,
  K = my_K_intial, init_pi = c(0.9, 0.1),
  niter = 10000, burnin = 1000, thin = 1, verbose = TRUE
)
```

```
Iter    1 | LL=-492.5 | 642.2 iter/sec | acc_rate_nu=[0.000 0.000]
Iter  500 | LL=-452.0 | 938.5 iter/sec | acc_rate_nu=[0.050 0.064]
Iter 1000 | LL=-453.3 | 962.1 iter/sec | acc_rate_nu=[0.048 0.061]
Iter 1500 | LL=-457.4 | 961.6 iter/sec | acc_rate_nu=[0.047 0.063]
Iter 2000 | LL=-452.7 | 971.8 iter/sec | acc_rate_nu=[0.046 0.065]
Iter 2500 | LL=-452.9 | 969.3 iter/sec | acc_rate_nu=[0.046 0.066]
```

```

Iter 3000 | LL=-456.8 | 973.1 iter/sec | acc_rate_nu=[0.047 0.067]
Iter 3500 | LL=-452.9 | 970.1 iter/sec | acc_rate_nu=[0.046 0.067]
Iter 4000 | LL=-451.8 | 974.7 iter/sec | acc_rate_nu=[0.044 0.067]
Iter 4500 | LL=-452.5 | 977.3 iter/sec | acc_rate_nu=[0.043 0.065]
Iter 5000 | LL=-452.2 | 977.4 iter/sec | acc_rate_nu=[0.043 0.062]
Iter 5500 | LL=-452.0 | 976.2 iter/sec | acc_rate_nu=[0.042 0.060]
Iter 6000 | LL=-454.2 | 976.3 iter/sec | acc_rate_nu=[0.042 0.059]
Iter 6500 | LL=-454.3 | 976.8 iter/sec | acc_rate_nu=[0.041 0.059]
Iter 7000 | LL=-455.6 | 976.6 iter/sec | acc_rate_nu=[0.042 0.059]
Iter 7500 | LL=-454.0 | 976.0 iter/sec | acc_rate_nu=[0.041 0.059]
Iter 8000 | LL=-451.0 | 975.2 iter/sec | acc_rate_nu=[0.041 0.060]
Iter 8500 | LL=-453.6 | 975.8 iter/sec | acc_rate_nu=[0.041 0.060]
Iter 9000 | LL=-452.8 | 976.6 iter/sec | acc_rate_nu=[0.041 0.061]
Iter 9500 | LL=-453.5 | 977.6 iter/sec | acc_rate_nu=[0.041 0.061]
Iter 10000 | LL=-453.1 | 976.9 iter/sec | acc_rate_nu=[0.040 0.061]

```

```
# posterior summaries
```

```
colMeans(fit$pi)
```

```
[1] 0.7891114 0.2108886
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.8 0.2
```

```
apply(fit$nu, 2, mean)
```

```
[1] 3.035037 10.195513
```

```
dat$nu # true nu
```

```
[1] 3 12
```

Check identifiability issue

```
# run the model again with swapped initial weights
```

```
set.seed(1234)
```

```
fit2 <- moewishart(S_list,
```

```
  K = my_K_intial, init_pi = c(0.1, 0.9),
```

```
  niter = 10000, burnin = 1000, thin = 1, verbose = TRUE
```

```
)
```

```

Iter    1 | LL=-492.5 | 891.3 iter/sec | acc_rate_nu=[0.000 0.000]
Iter   500 | LL=-451.0 | 932.4 iter/sec | acc_rate_nu=[0.040 0.060]
Iter  1000 | LL=-453.9 | 953.2 iter/sec | acc_rate_nu=[0.038 0.056]
Iter  1500 | LL=-455.2 | 959.7 iter/sec | acc_rate_nu=[0.038 0.057]
Iter  2000 | LL=-452.9 | 958.0 iter/sec | acc_rate_nu=[0.041 0.052]
Iter  2500 | LL=-456.3 | 965.9 iter/sec | acc_rate_nu=[0.042 0.054]
Iter  3000 | LL=-453.9 | 971.3 iter/sec | acc_rate_nu=[0.042 0.053]
Iter  3500 | LL=-458.2 | 975.2 iter/sec | acc_rate_nu=[0.041 0.053]
Iter  4000 | LL=-453.8 | 975.9 iter/sec | acc_rate_nu=[0.041 0.056]
Iter  4500 | LL=-452.8 | 977.1 iter/sec | acc_rate_nu=[0.042 0.056]
Iter  5000 | LL=-453.4 | 975.1 iter/sec | acc_rate_nu=[0.042 0.059]
Iter  5500 | LL=-452.1 | 975.7 iter/sec | acc_rate_nu=[0.044 0.057]
Iter  6000 | LL=-453.3 | 973.4 iter/sec | acc_rate_nu=[0.043 0.058]
Iter  6500 | LL=-453.2 | 972.2 iter/sec | acc_rate_nu=[0.042 0.056]
Iter  7000 | LL=-456.5 | 973.1 iter/sec | acc_rate_nu=[0.043 0.055]

```

```

Iter 7500 | LL=-452.7 | 973.8 iter/sec | acc_rate_nu=[0.044 0.056]
Iter 8000 | LL=-453.0 | 974.2 iter/sec | acc_rate_nu=[0.043 0.056]
Iter 8500 | LL=-456.8 | 973.9 iter/sec | acc_rate_nu=[0.043 0.056]
Iter 9000 | LL=-456.1 | 974.2 iter/sec | acc_rate_nu=[0.043 0.057]
Iter 9500 | LL=-452.0 | 974.6 iter/sec | acc_rate_nu=[0.042 0.057]
Iter 10000 | LL=-452.0 | 972.0 iter/sec | acc_rate_nu=[0.043 0.057]

```

```
# posterior summaries
```

```
colMeans(fit2$pi)
```

```
[1] 0.2090747 0.7909253
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.8 0.2
```

```
apply(fit2$nu, 2, mean)
```

```
[1] 10.643540 3.012158
```

```
dat$nu # true nu
```

```
[1] 3 12
```

Conclusion:

The two model fittings above with initial weights `init_pi=c(0.9, 0.1)` and `init_pi=c(0.1, 0.9)` result in similar results, which indicate possibly identifiability issue.

It might be NOT possible based on the proof of Proposition 2 in Yakowitz and Spragins (1968) to give similar proofs to show identifiability of the MoE of (inverse) Wishart distributions.

EM algorithm

```
# run the model with EM algorithm
```

```
set.seed(1234)
```

```
fit3 <- moewishart(S_list, method = "em",
  K = my_K_initial, init_pi = c(0.1, 0.9),
  niter = 10000, burnin = 1000, thin = 1, verbose = TRUE
)
```

```

Iter 10 | Loglik: -1542.2885 | Nu: 3.4, 3.14
Iter 20 | Loglik: -1515.2390 | Nu: 2.99, 6.13
Iter 30 | Loglik: -1510.3904 | Nu: 2.98, 8.52
Iter 40 | Loglik: -1509.8677 | Nu: 2.98, 9.45
Iter 50 | Loglik: -1509.8129 | Nu: 2.98, 9.77
Iter 60 | Loglik: -1509.8070 | Nu: 2.98, 9.87
Iter 70 | Loglik: -1509.8063 | Nu: 2.98, 9.91
Iter 80 | Loglik: -1509.8062 | Nu: 2.98, 9.92
Converged at iteration 84

```

```
# EM estimates
```

```
fit3$pi
```

```
[1] 0.7926627 0.2073373
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.8 0.2
```

```
fit3$nu
```

```
[1] 2.981787 9.919218
```

```
dat$nu # true nu
```

```
[1] 3 12
```